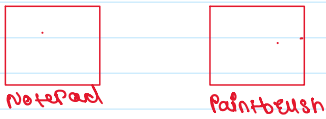
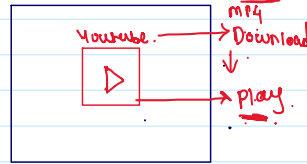


IPC - Interprocess communication.

Independent process

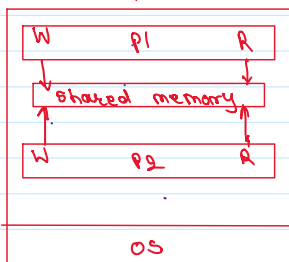


Co-operative process

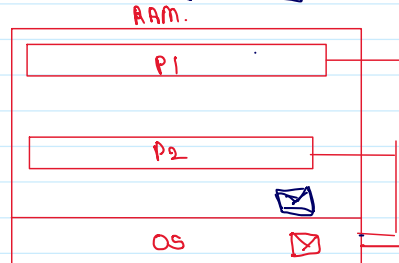


IPC mechanisms. ① shared memory ② message passing

shared memory



message passing



message passing models - pipe, signals, message queue, socket.

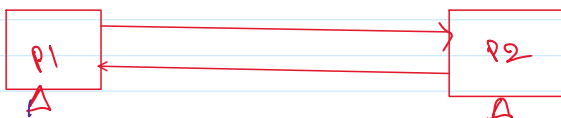
Pipe



- ① unidirectional ipc mech.
- ② stream based mech.
- ③ terminal > cmd1 | cmd2

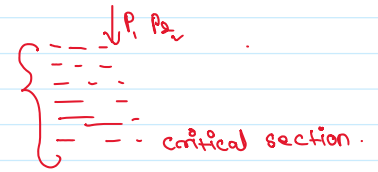
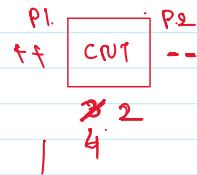
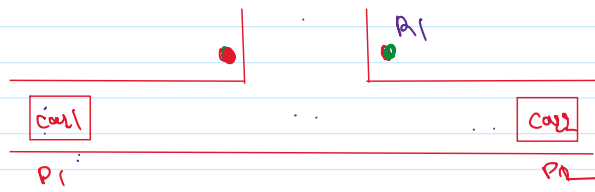
output
 input

Socket (communication end points).



Race condition - when multiple processes are trying to access a

Race condition - when multiple processes are trying to access a resource



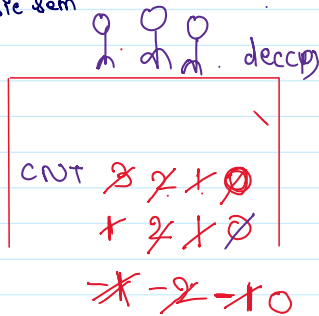
Peterson's Problem

synchronization objects = ① semaphore
② mutex.

Semaphore:- It is proposed by Dijkstra.

- It is Counter.
- decrement / wait op / pop.
- dec sema CNT by 1
- if sema cnt < 0 block one of the process.
- increment / signal op / V op.
- inc sema CNT by 1
- if one or more processes block wake up one of the process

Counting sem / classic sem

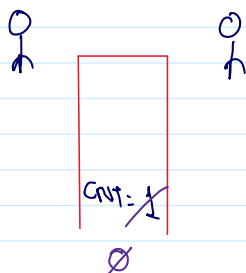


Counting sem / classic sem - when multiple processes are trying to access a resource.

binary sem - when a process acquires a resource

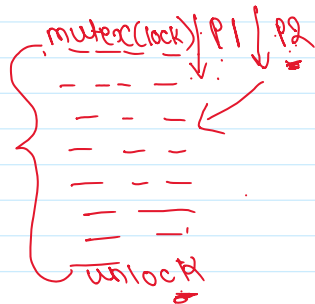
When sem CNT = n block processes n.

binary sem



1
0

mutex - when a process acquired a resource.



Deadlock.

